



Urban Flood Monitoring and Visualization System

Based on Citizen Science

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1. INTRODUCTION

Urban flooding is a recurring problem in the Metropolitan Area of Asunción, affecting mobility, safety, and quality of life. The lack of localized, real-time information hinders the identification of critical areas and timely decision-making. In this context, citizen science enables community-generated reports to serve as a complementary information source for monitoring flood events.

Objective: To develop a web platform for collecting, visualizing, and managing citizen reports related to urban flooding, integrating georeferenced information, visual evidence, and descriptive event data.

2. METHODOLOGY

Heavy rainfall can lead to flood events that are documented by citizens through photographs and georeferenced reports. The collected information is then integrated into the platform and visualized on an interactive map.



Figure 1. General workflow of the proposed urban flood monitoring system.

Study area: The system focuses on the Metropolitan Area of Asunción, an urban region exposed to recurrent flood and waterlogging events.

Data sources: The platform integrates two main sources: georeferenced citizen reports and historical news records on flood events. Citizen reports include GPS location, event description, observed impact, report status, and an optional photograph. Historical records provide complementary data to contextualize past events and support spatial analysis.

System architecture: Information is submitted through a Next.js, React, and TypeScript web application, validated by the backend, and stored in Supabase using PostgreSQL and Storage. Data is then displayed on an interactive geospatial map to support event consultation, identify affected areas, and monitor urban flooding.

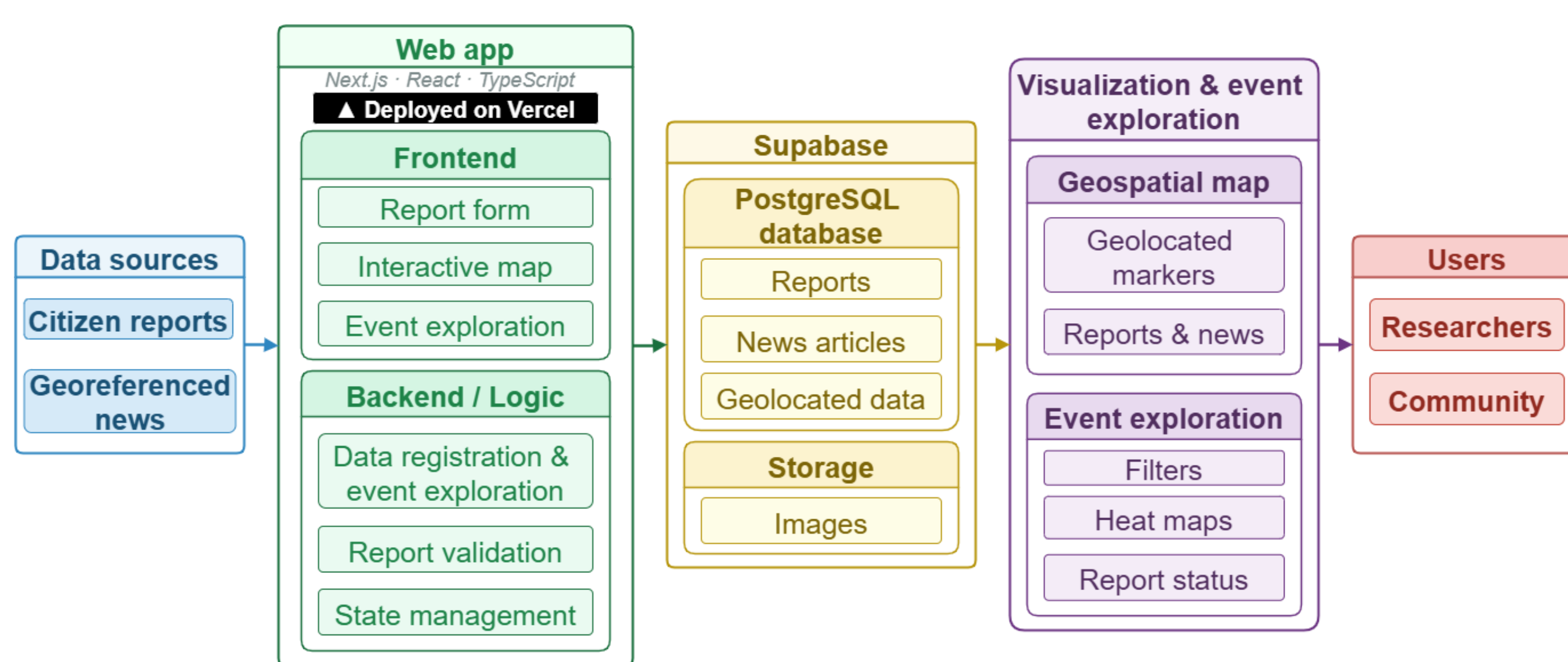


Figure 2. System architecture of the urban flood monitoring and visualization platform based on citizen science.

3. RESULTS

The results show the implementation of a citizen science platform that integrates georeferenced citizen reports and historical news to support urban flood monitoring. The platform provides tools for data collection, visualization, filtering, and analysis, promoting community participation and generating valuable information for decision-making.

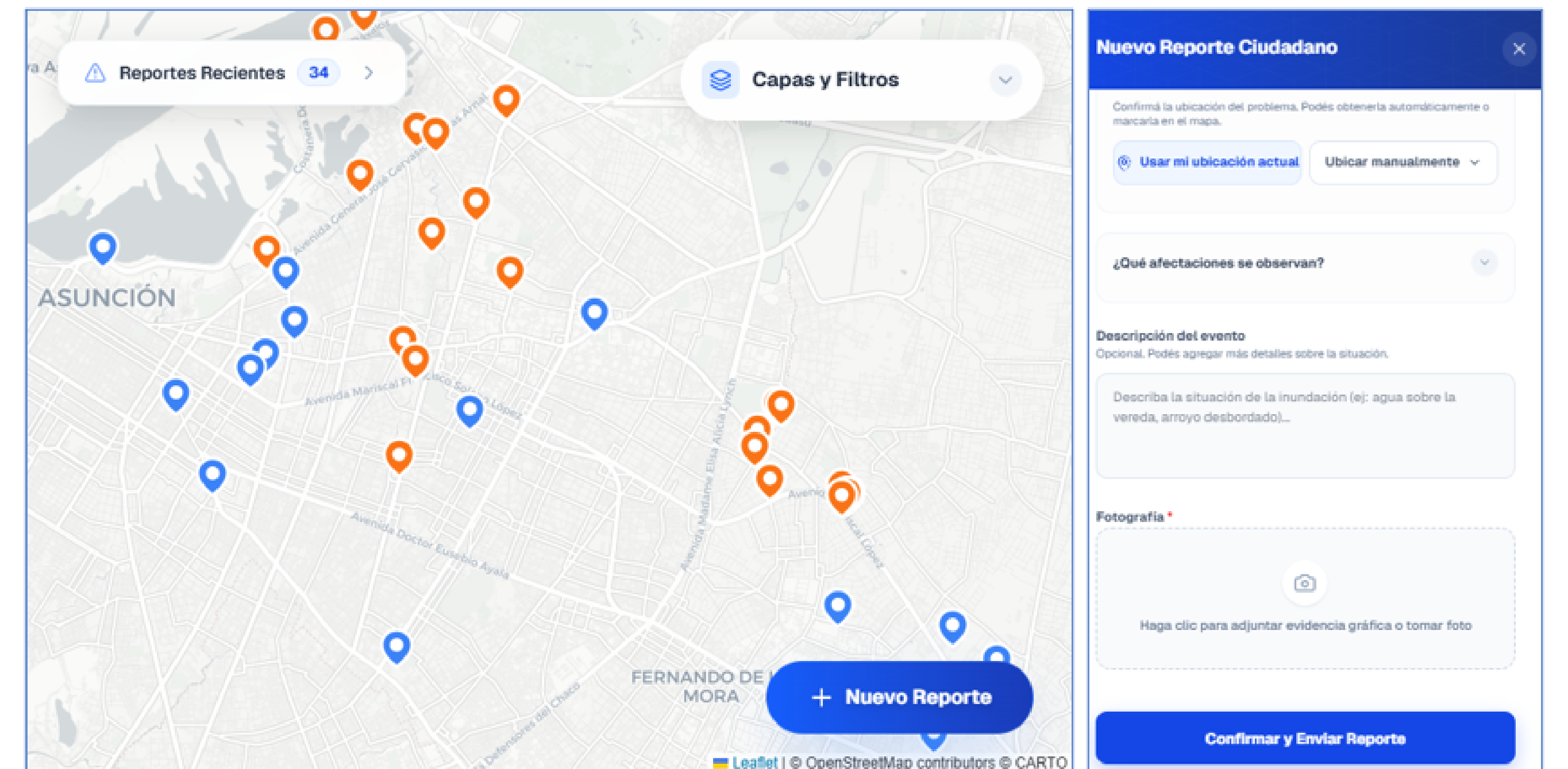


Figure 3. Interactive map and web form for the registration of citizen reports.

The system allows users to submit new flood reports through a web form that includes geographic location, event description, and photographs.



Figure 4. Events displayed on the platform: citizen report and historical news.

The reported events are displayed on the map using interactive markers that show descriptive information and images associated with citizen reports and historical news.

4. CONCLUSION

The developed platform allows the collection and visualization of georeferenced information about urban flood events through the integration of citizen reports and historical news. The results show that citizen science can be used to support flood monitoring and identify affected areas in the Metropolitan Area of Asunción. Future work includes the incorporation of IoT sensors, automated report validation mechanisms, and spatial analysis tools to identify critical flood zones and strengthen urban monitoring capabilities.

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REFERENCES

- [1] Rick Bonney, Caren B. Cooper, Janis Dickinson, Steve Kelling, Tina Phillips, Kenneth V. Rosenberg, and Jennifer Shirk. "Citizen science: A developing tool for expanding science knowledge and scientific literacy". *BioScience*, 2009.
- [2] Muki Haklay. "Citizen science and volunteered geographic information: Overview and typology of participation". In: *Crowdsourcing Geographic Knowledge*, 2013.
- [3] SITI OSS. Open source software for situational intelligence.